

Colour Segmentation Using MATLAB

PANKAJ RAUT

This is a program for colour separation in an image. The program can be modified for a specific application, such as colour

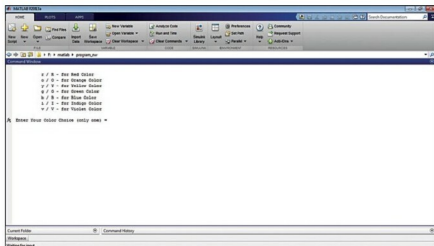


Fig. 1: Screenshot of the program output for colour selection

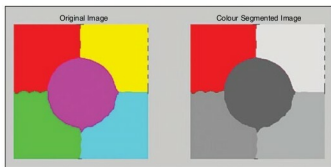


Fig. 2: Program output with red colour image

EFY Note

The source code of this project is included in this month's EFY DVD and is also available for free download at source.efymag.com

detection in processing industry (food, textile, etc) or robotic vision in automation industry.

The program was developed using MATLAB version 7.7 (or R2008b). MATLAB was used because it has digital image processing (DIP) tool which

is very easy to implement in this project.

Colour is related to light. Human eye perceives light as electromagnetic radiation with wavelengths approximately between 380nm and

760nm. The visible colour spectrum has violet, indigo, blue, green, yellow, orange and red (VIBGYOR) colours.

Approximate VIBGYOR colour wavelengths in nanometres (nm) are given below:

Colour	Wavelength (in nm)
Red	720-760
Orange	590-620
Yellow	545-590
Green	490-525
Blue	450-490
Indigo	420-450
Violet	380-420

The main application of this program is to separate each colour from the entire VIBGYOR colour spectrum in an image.

If you have an application that requires a specific colour detection or separation, this program would come handy for you. For example, if you want red colour from an image, run this program. It will prompt you to import an image into the screen. After importing the image, you need to select the first letter of the colour (see Fig. 1) to be separated. Select R option for red. You will get only red colour and rest of the image will be in various shades of grey, as shown in Fig. 2.

The advantage of this program is that it can separate each colour from the seven colours instantly.

Software

Some important functions used in the code are described below.

The code starts with the following function declaration:

```
function C = VIBGYORsegmentation(img)
```

User has to import an image in .jpg format. This is achieved by the following code:

```
[filename,pathname]=uigetfile('.jpg',  
'select an image');
```

Note that, the image imported by the user should be a colour image. The program outputs two images on the screen. The first image on left side of the screen is the original and the second on the right side is the colour-segmented image. The following codes are used to achieve this:

```
figure, subplot(1,2,1),imshow(uint8  
(img),[]);title('Original Image');  
subplot(1,2,2),imshow(uint8(C),[]);  
title('Color Segmented Image');
```

The author is a B.Tech (electronics and communication) student of Jawaharlal Nehru Technological University, Hyderabad. His interests include computer vision, sensors and electronics, social computing, robotics, optics and MATLAB